2024

# Machine Learning - 2

PGP-DSBA Online

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**Context: Problem 1:**

You are in discussions with ABC Consulting Company for providing transport for their employees. For this purpose, you are tasked with understanding how do the employees of ABC Consulting prefer to commute presently (between home and office). Based on the parameters like age, salary, work experience etc. given in the data set 'Transport.csv', you are required to predict the preferred mode of transport. The project requires you to build several Machine Learning models and compare them so that the model can be finalized.

**Objective:**

The objective is to build various Machine Learning models on this data set and based on the accuracy metrics decide which model is to be finalized for finally predicting the mode of transport chosen by the employee.

**Data Dictionary:**

- **Age**: Age of the Employee in Years
- **Gender**: Gender of the Employee
- **Engineer**: For Engineer =1, Non Engineer =0
- **MBA**: For MBA =1 , Non-MBA =0
- **Work Exp**: Experience in years
- **Salary**: Salary in Lakhs per Annum
- **Distance**: Distance in km from Home to Office
- **license**: If Employee has Driving License -1, If not, then 0
- **Transport**: Mode of Transport

**Data Analysis & A detailed solution around the problems:**

**Overview of the dataset:** The initial steps to get an overview of any dataset are to:

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not.
- Get information about the number of rows and columns in the dataset.
- Find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

**Sample of the dataset:**

| | Age | Gender | Engineer | MBA | Work Exp | Salary | Distance | license | Transport |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 28 | Male | 0 | 0 | 4 | 14.3 | 3.2 | 0 | Public Transport |
| **1** | 23 | Female | 1 | 0 | 4 | 8.3 | 3.3 | 0 | Public Transport |
| **2** | 29 | Male | 1 | 0 | 7 | 13.4 | 4.1 | 0 | Public Transport |
| **3** | 28 | Female | 1 | 1 | 5 | 13.4 | 4.5 | 0 | Public Transport |
| **4** | 27 | Male | 1 | 0 | 4 | 13.4 | 4.6 | 0 | Public Transport |

**Observation 1:** The dataset has 444 rows and 9 columns.

**Observation 2:** All the columns in the data are non-null. We have Gender & Transport in categorical (object) type, rest others in numerical (integer) type.

**Observation 3:** It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are no missing values in the data, and we are good to proceed ahead.

**Observation 4:** There are 444 unique employee data. There are no duplicate values in the data. We observe 2 modes of transport – Public & Private.

**Observation 5:** Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Age** | 444.0 | 27.747748 | 4.416710 | 18.0 | 25.0 | 27.0 | 30.000 | 43.0 |
| **Engineer** | 444.0 | 0.754505 | 0.430866 | 0.0 | 1.0 | 1.0 | 1.000 | 1.0 |
| **MBA** | 444.0 | 0.252252 | 0.434795 | 0.0 | 0.0 | 0.0 | 1.000 | 1.0 |
| **Work Exp** | 444.0 | 6.299550 | 5.112098 | 0.0 | 3.0 | 5.0 | 8.000 | 24.0 |
| **Salary** | 444.0 | 16.238739 | 10.453851 | 6.5 | 9.8 | 13.6 | 15.725 | 57.0 |
| **Distance** | 444.0 | 11.323198 | 3.606149 | 3.2 | 8.8 | 11.0 | 13.425 | 23.4 |
| **license** | 444.0 | 0.234234 | 0.423997 | 0.0 | 0.0 | 0.0 | 0.000 | 1.0 |

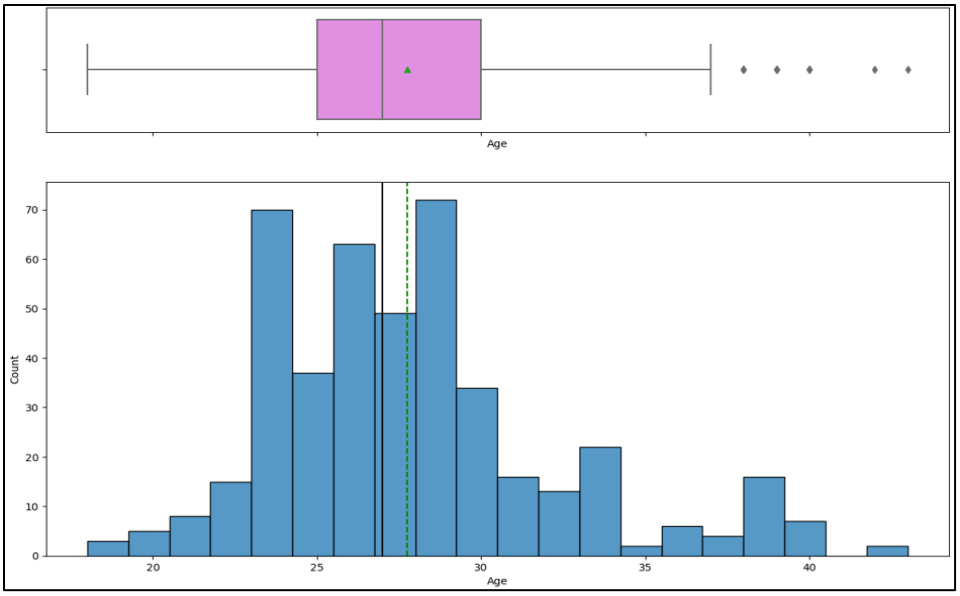
## Topic 1: Exploratory Data Analysis:

Let's check the count of each unique category in each of the categorical variables:

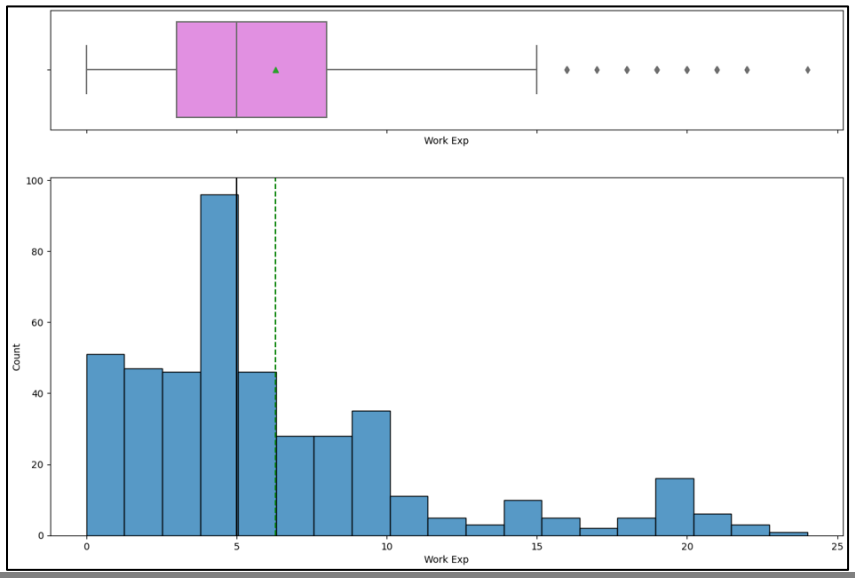
```
Male      316
Female    128
Name: Gender, dtype: int64
-----------------------------------------------
Public Transport     300
Private Transport    144
Name: Transport, dtype: int64
-----------------------------------------------
```

**1. Univariate analysis: (The green dotted lines in all the image denotes average of the data)**

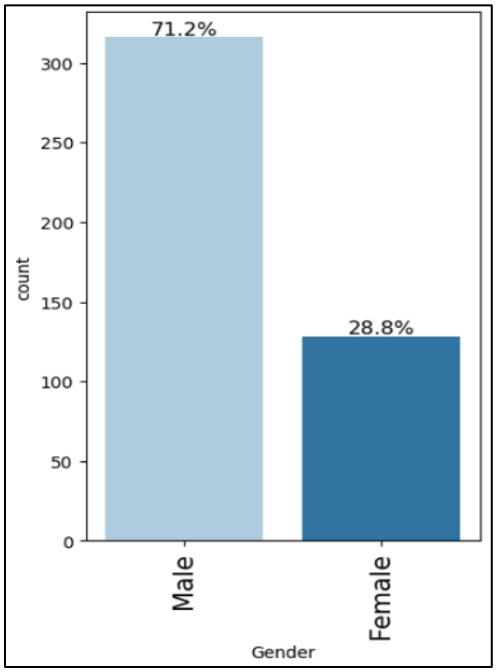
**1.1 Observation on Age:** Data is quite spread; we observe few outliers, and an average age of 28.



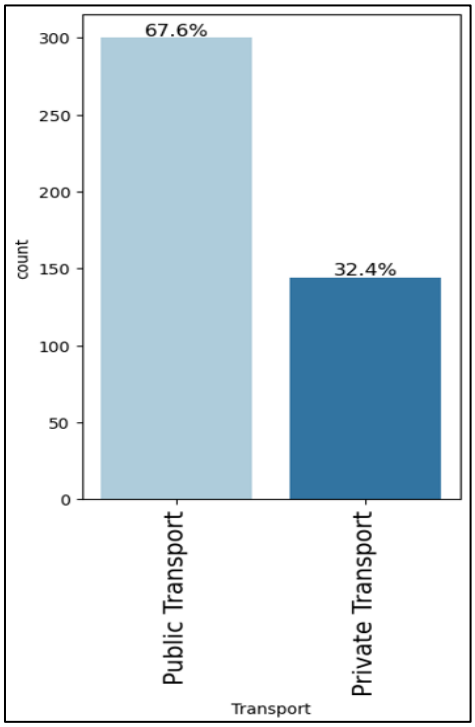

**1.2 Observation on Work Experience:** Data is right skewed, with outliers observed, and an average experience of 6 years.



**1.3 Observation on Gender:** We observe 71% data as male employees and 29% as female employees.

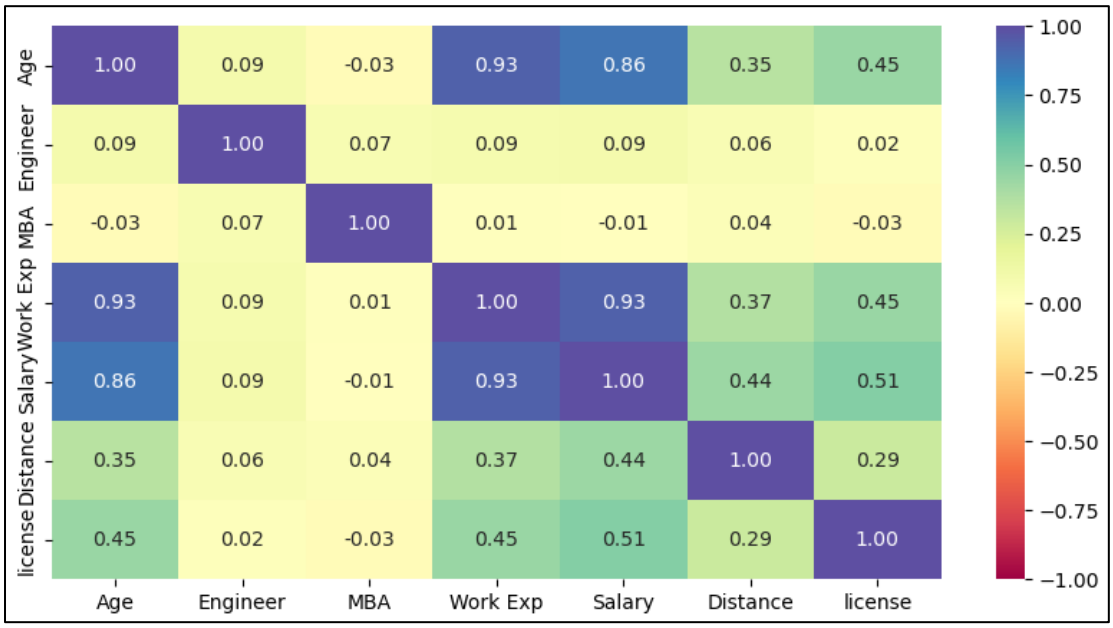



**1.4 Observation on Transport:** Public transport is clearly preferred more with 68% as compared to private transport with 32%.



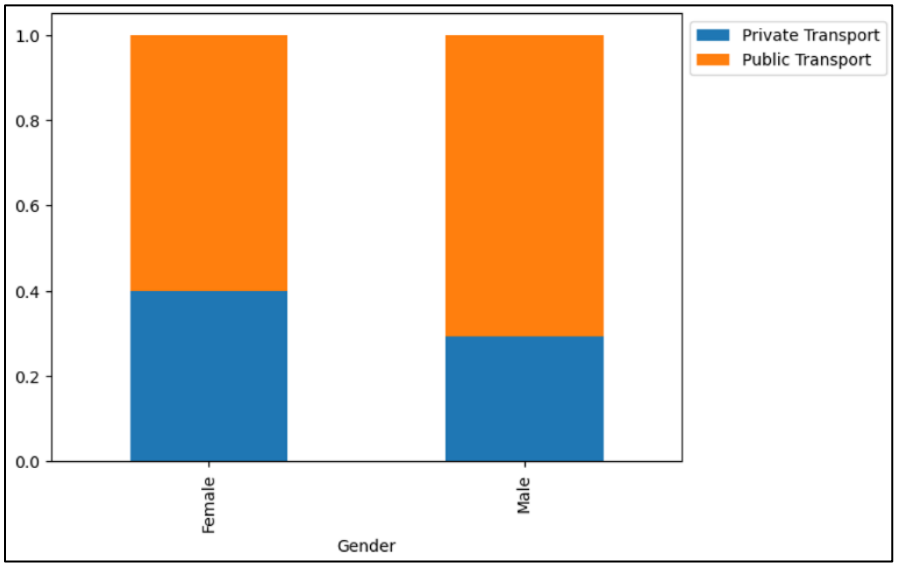
**2. Bivariate Analysis: Let's check for correlations:**

**2.1 Heatmap:** We see high correlation between age & work experience, followed by work experience & salary, with a decent correlation between salary & age.



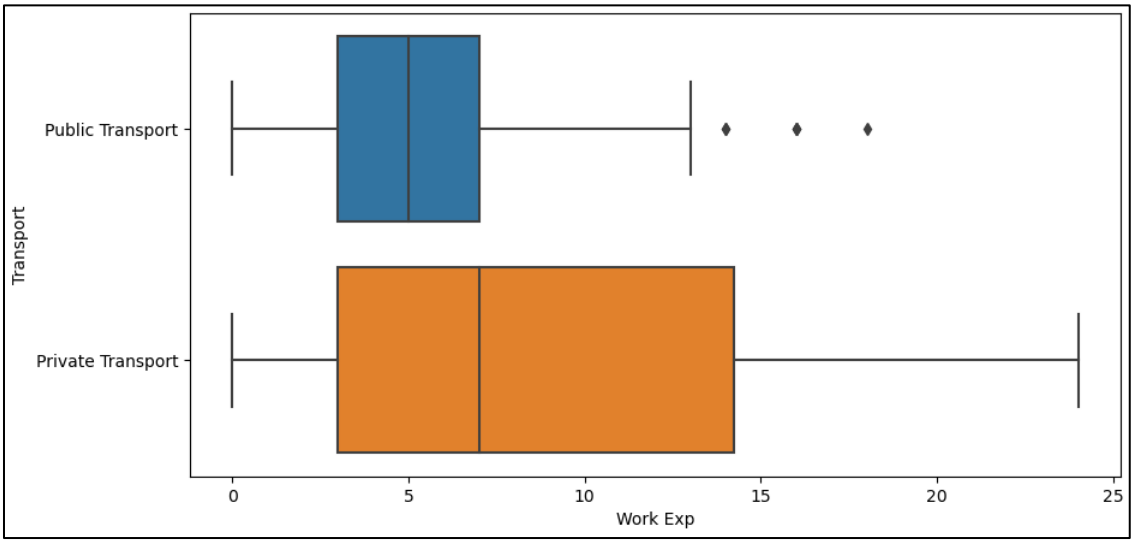

**2.2 Let's find out if Gender has any impact on Transport Preference:** We see that female employees prefer private transport more as compared to male using private transport. And male preferring public transport 70% of the time as compared to female employees.

| Transport | Private Transport | Public Transport | All |
|---|---|---|---|
| Gender | | | |
| All | 144 | 300 | 444 |
| Male | 93 | 223 | 316 |
| Female | 51 | 77 | 128 |



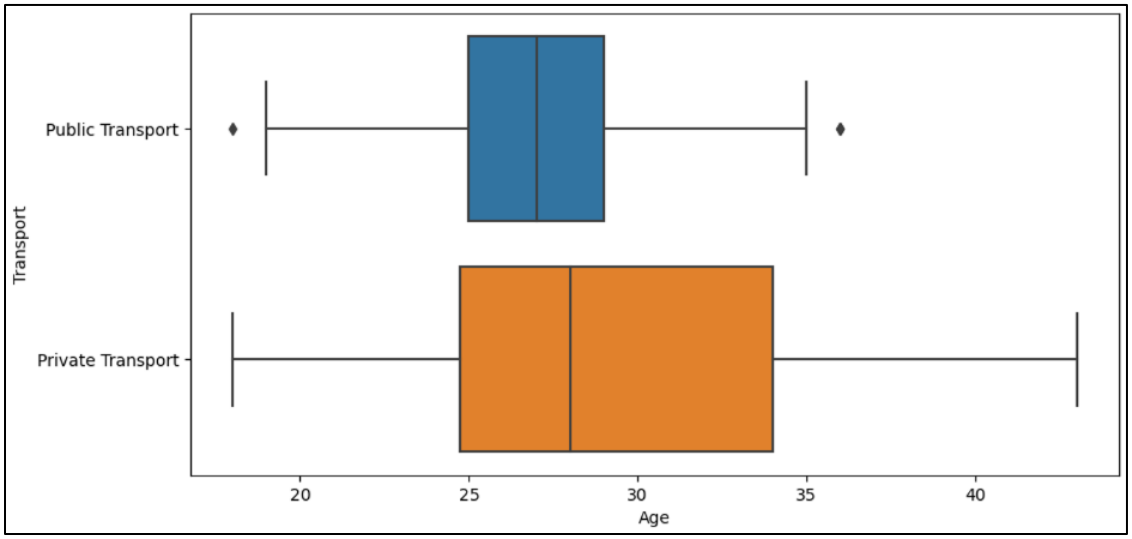
**2.3 Checking if the Work Experience of the employee has any relation with preference of the transport:**

- We observe few outliers in public transport; we see public transport being used 77% times by employees with work experience from 2 to 8 years.
- We see private transport being quite less used overall which is 32% only and often used by employees with work experience ranging from 2 to 14 years with 62% times.



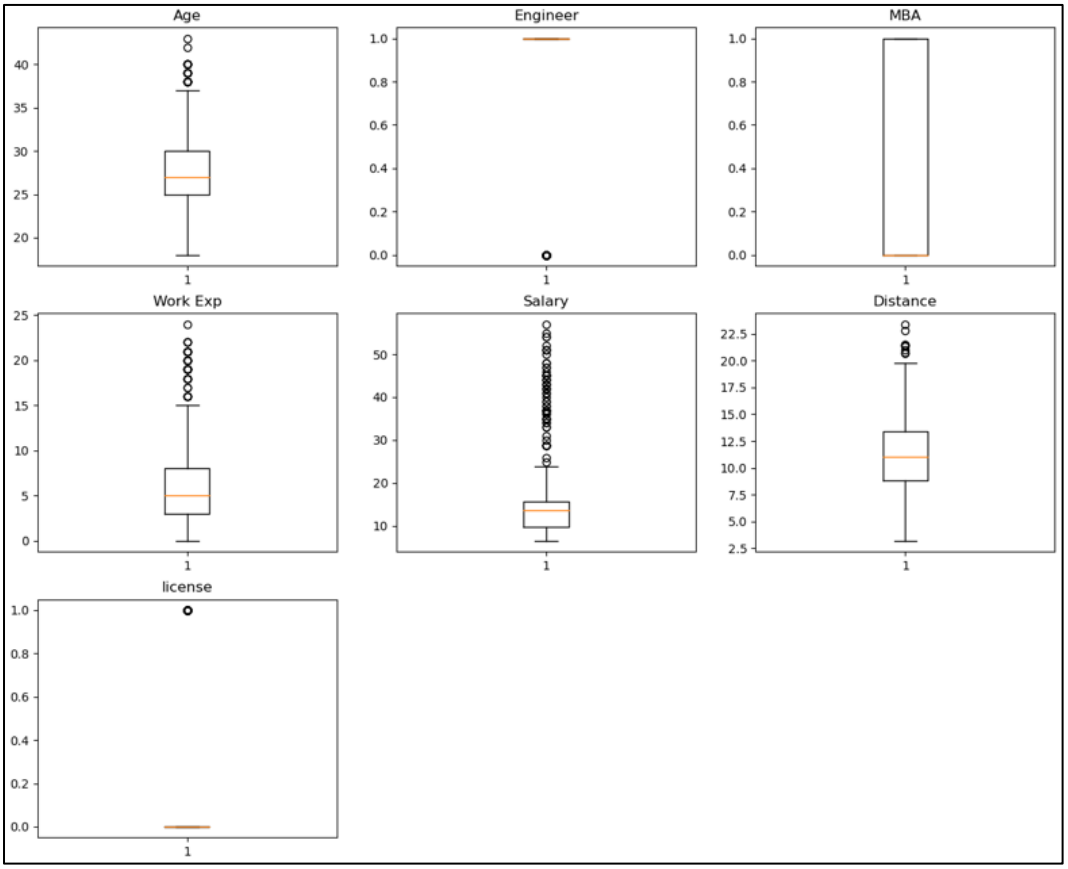

**2.4 Checking if the Age of the employee has any relation with preference of the Transport:**

- We notice private transport being mostly used by age group 22-28.
- We see public transport being highly used by age group 24 to 30, which is 80% times.



## Topic 2: Data Pre-processing:

1. **Outlier check:** We observe outliers in most of the features, with too many outliers in Salary.




2. **Data preparation for modeling:** Before we proceed to build a model, we'll have to encode categorical features. We'll split the data into train and test to be able to evaluate the model that we build on the train data.

```
Shape of Training set :  (310, 8)
Shape of test set :  (134, 8)
Percentage of classes in training set:
1    0.674194
0    0.325806
Name: Transport, dtype: float64
Percentage of classes in test set:
1    0.679104
0    0.320896
Name: Transport, dtype: float64
```

## Topic 3: Model Evaluation Criterion:

**Model can make wrong predictions as**:

1. Model predicts that the Public mode of transport is preferred but employee prefers private mode.
2. Model predicts that that the Private mode of transport is preferred but employee prefers public mode.

**Which case is more important?**

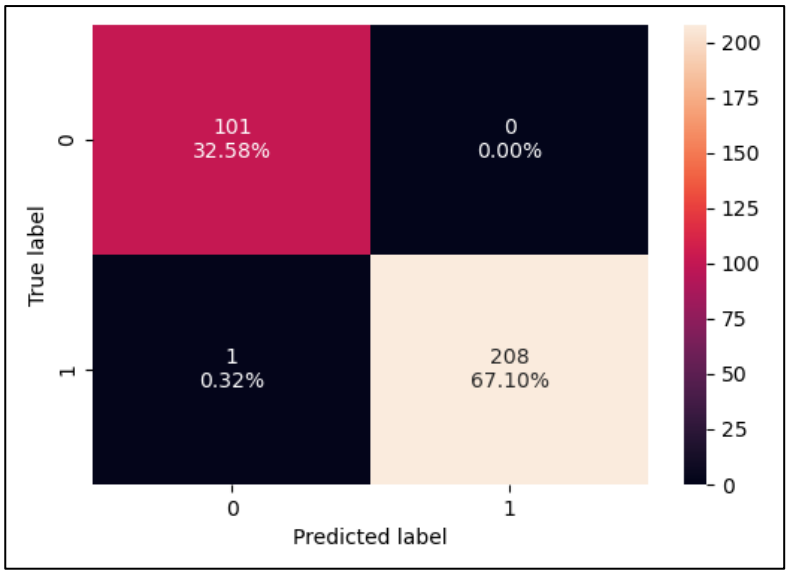
Both are important to correctly estimate the number of employees preferring the private transport

**How to reduce the losses?**

- **F1 Score** can be used as the metric for evaluation of the model, greater the F1 score higher are the chances of minimizing False Negatives and False Positives.
- We will use balanced class weights so that model focuses equally on both classes.

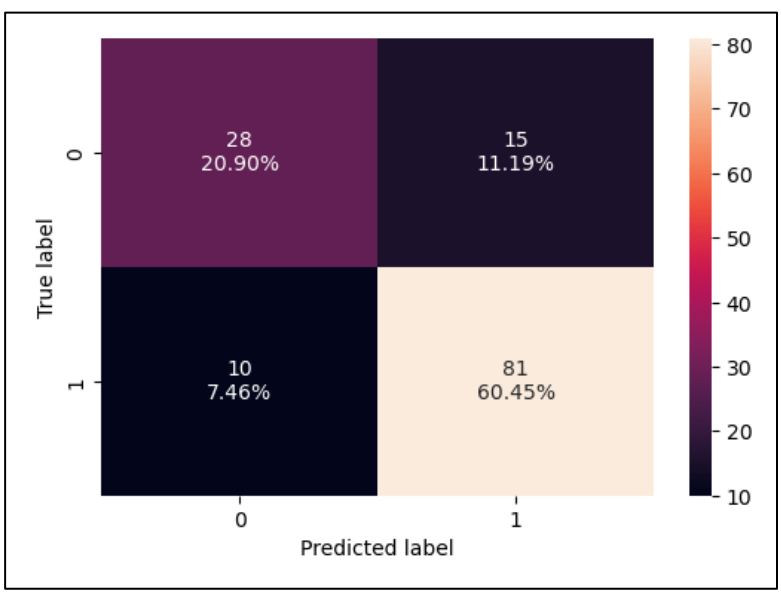
**<u>Topic 4: Bagging - Model Building and Hyper-parameter Tuning:</u>** Bagging is an ensemble method where multiple instances of a base learning algorithm are trained on different subsets of the training data.

1. **Bagging Classifier:** It helps reduce over fitting and variance by combining the predictions of multiple models trained on different subsets of data.
    - Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.996774 | 0.995215 | 1.0 | 0.997602 |

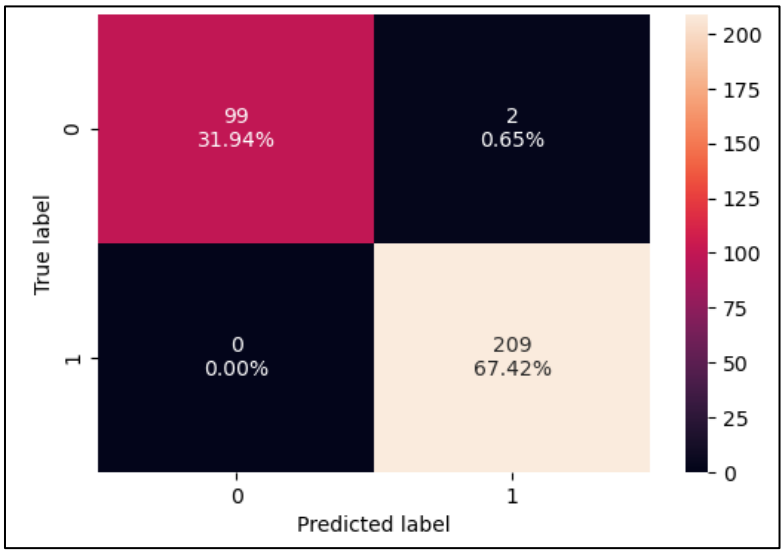
- Checking model performance on test set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.813433 | 0.89011 | 0.84375 | 0.86631 |

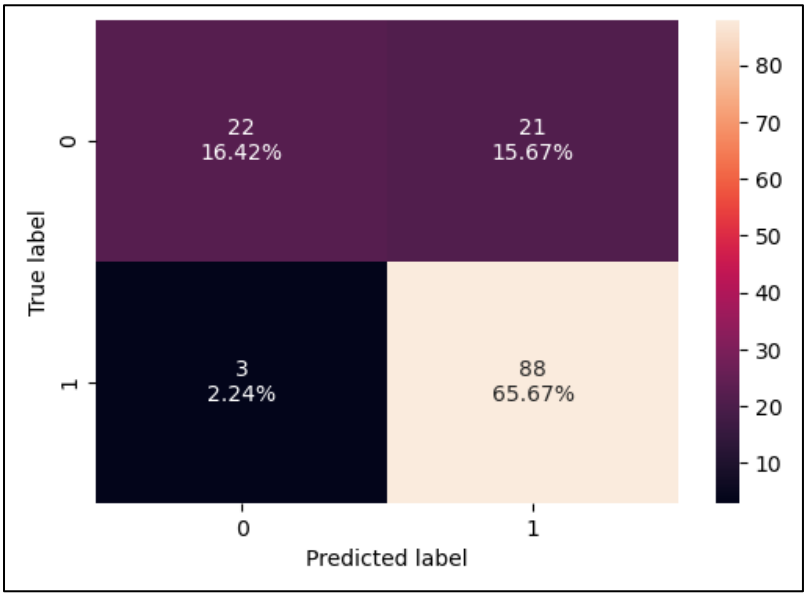
**2. Hyper-parameter Tuning - Bagging Classifier:**

- Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.993548 | 1.0 | 0.990521 | 0.995238 |

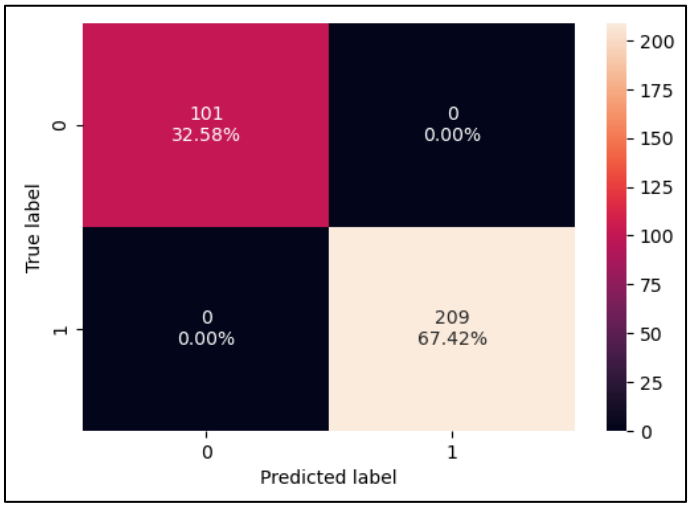
- Checking model performance on test set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.820896 | 0.967033 | 0.807339 | 0.88 |

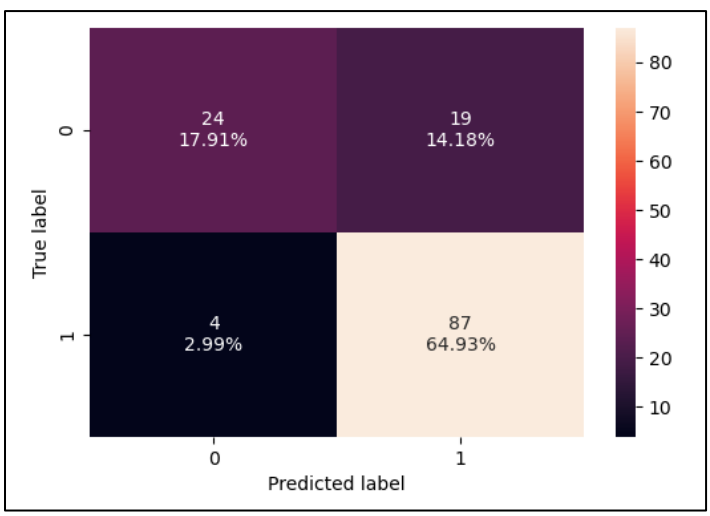
3. **Random Forest:** Random Forest is a popular ensemble learning algorithm based on bagging. It builds multiple decision trees during training and averages their predictions to improve accuracy and robustness.

- Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 1.0 | 1.0 | 1.0 | 1.0 |

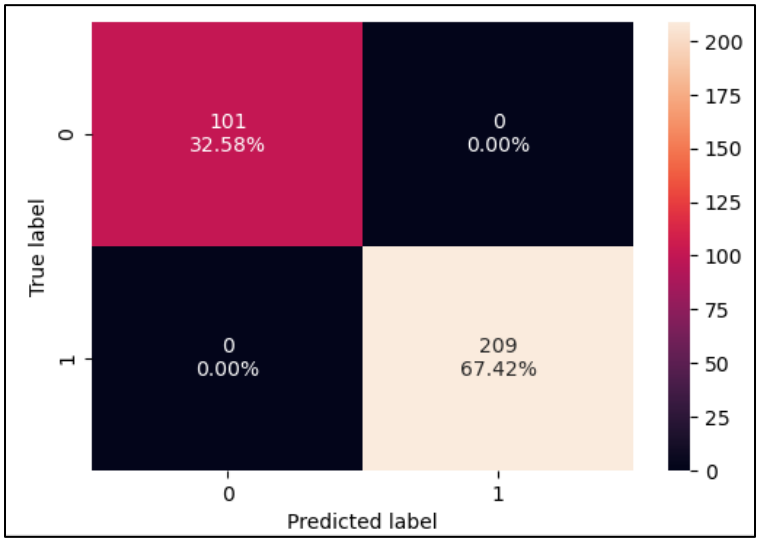
- Checking model performance on test set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.828358 | 0.956044 | 0.820755 | 0.883249 |

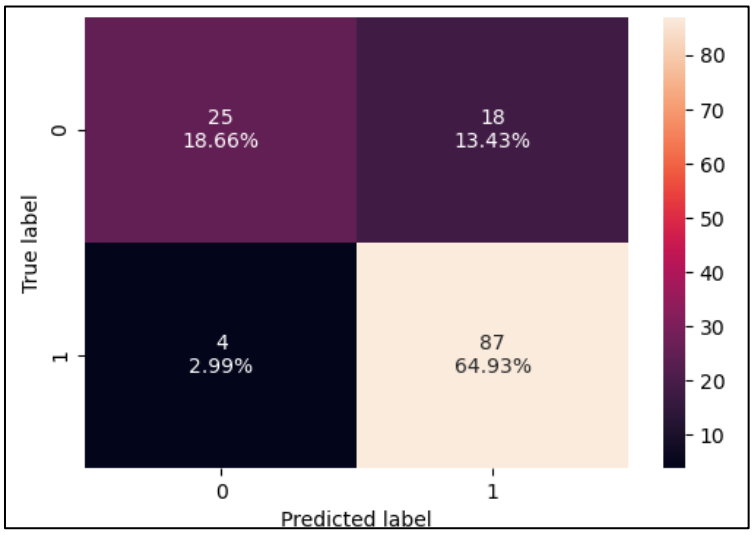
**4. Hyper-parameter Tuning – Random Forest:**

- Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 1.0 | 1.0 | 1.0 | 1.0 |

- Checking model performance on test set:

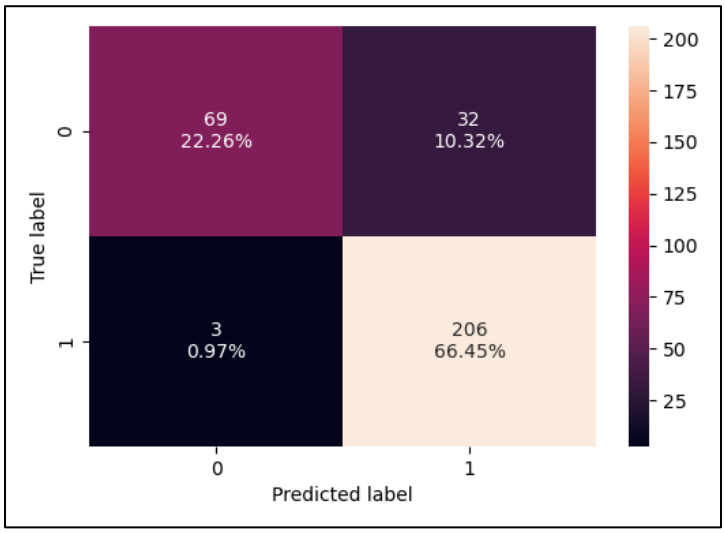



| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.835821 | 0.956044 | 0.828571 | 0.887755 |

**Topic 5: Boosting - Model Building and Hyper-parameter Tuning:** Boosting algorithms aim to reduce bias and increase model performance by iteratively improving the model's predictive capability.

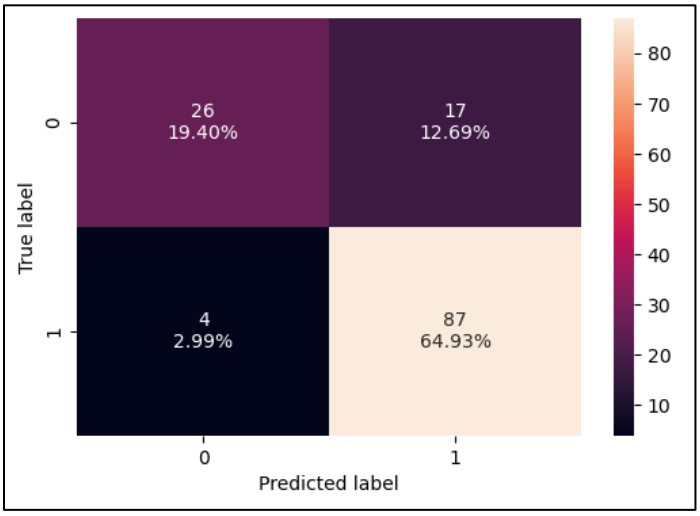
1. **AdaBoost Classifier:**

- Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.887097 | 0.985646 | 0.865546 | 0.9217 |

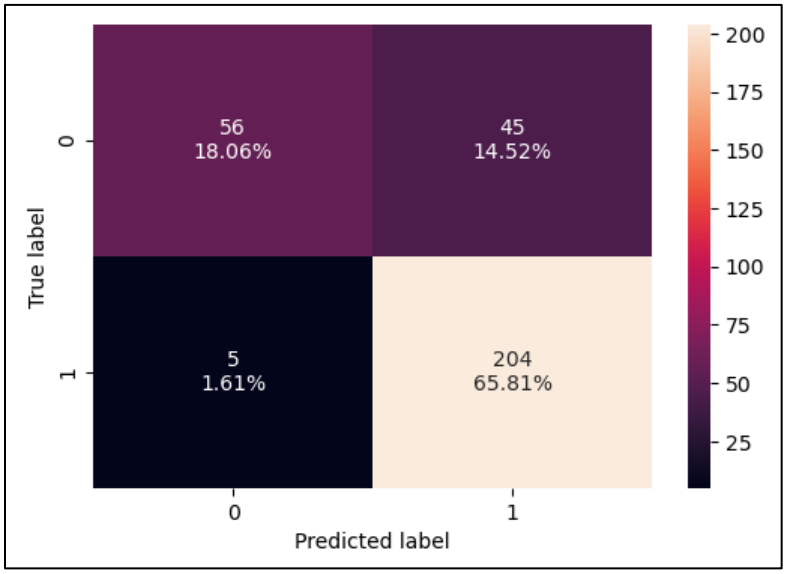
- Checking model performance on test set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.843284 | 0.956044 | 0.836538 | 0.892308 |

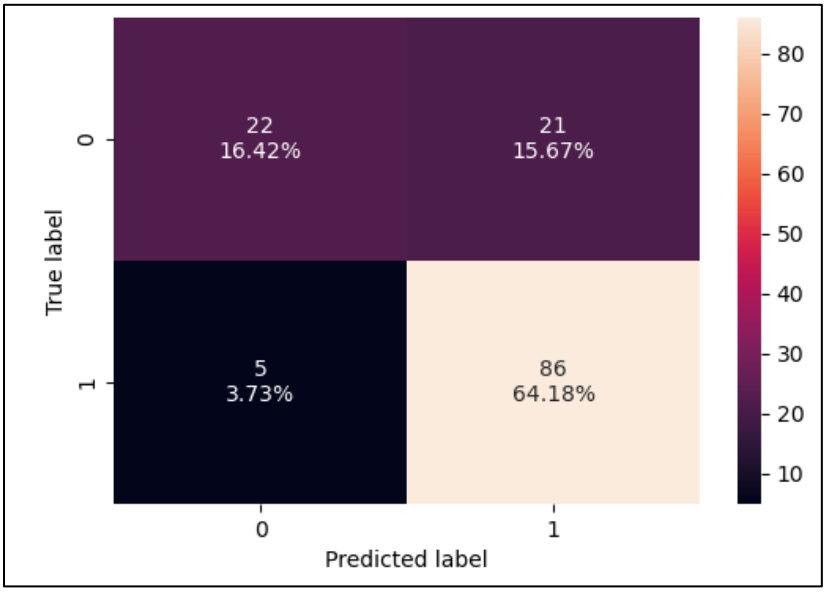
**2. Hyper-parameter Tuning - AdaBoost Classifier:**

- Checking model performance on training set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.83871 | 0.976077 | 0.819277 | 0.89083 |

- Checking model performance on test set:




| | Accuracy | Recall | Precision | F1 |
|---|---|---|---|---|
| **0** | 0.80597 | 0.945055 | 0.803738 | 0.868687 |

## Topic 6: Model Comparison and Final Model Selection:

### 1. Training performance comparison:

Training performance comparison:

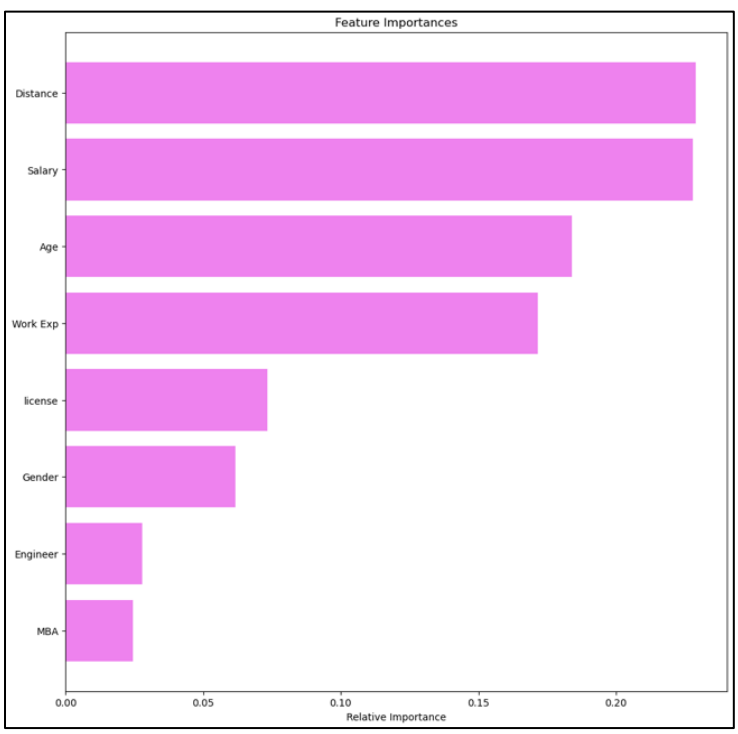
| | Bagging Classifier | Tuned Bagging Classifier | Random Forest | Tuned Random Forest | Adaboost Classifier | Tuned Adaboost Classifier |
|---|---|---|---|---|---|---|
| Accuracy | 0.996774 | 0.993548 | 1.0 | 1.0 | 0.887097 | 0.838710 |
| Recall | 0.995215 | 1.000000 | 1.0 | 1.0 | 0.985646 | 0.976077 |
| Precision | 1.000000 | 0.990521 | 1.0 | 1.0 | 0.865546 | 0.819277 |
| F1 | 0.997602 | 0.995238 | 1.0 | 1.0 | 0.921700 | 0.890830 |

### 2. Test performance comparison:

Test performance comparison:

| | Bagging Classifier | Tuned Bagging Classifier | Random Forest | Tuned Random Forest | Adaboost Classifier | Tuned Adaboost Classifier |
|---|---|---|---|---|---|---|
| Accuracy | 0.813433 | 0.820896 | 0.828358 | 0.835821 | 0.843284 | 0.805970 |
| Recall | 0.890110 | 0.967033 | 0.956044 | 0.956044 | 0.956044 | 0.945055 |
| Precision | 0.843750 | 0.807339 | 0.820755 | 0.828571 | 0.836538 | 0.803738 |
| F1 | 0.866310 | 0.880000 | 0.883249 | 0.887755 | 0.892308 | 0.868687 |

### 3. Important features of the model:



**Actionable Insights:**

- We could see that employees who are at lesser distance from Home to Office prefer Public Transport than employees who are farther away, who prefer Private Transport.

- Employees who have higher experience, in turn has more salary prefer to travel more through their own/private transport rather than taking public transport.

- We have high positive correlation between Age, Work Experience and Salary variables, which in turn is in positive relationship with predictor variable.

- We also see that male employees have license and female employees seem to use Public Transport, as they don't have license.

**Business Recommendations:**

- Distance and Salary are the two major factors followed by Age and Work Experience, which can be considered while predicting mode of transport for employees at ABC consulting.

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**Context: Problem 2:**

A dataset of Shark Tank episodes is made available. It contains 495 entrepreneurs making their pitch to the VC sharks. You will ONLY use "Description" column for the initial text mining exercise.

**Data Description:**

- Deal column suggests the companies secured business deal or not from the Sharks.
- Description suggests what pitch the entrepreneurs presented to the VC sharks.

**Data Analysis & A detailed solution around the problems:**

**Overview of the dataset:** The initial steps to get an overview of any dataset are to:

- Observe the first few rows of the dataset, to check whether the dataset has been loaded properly or not.
- Get information about the number of rows and columns in the dataset.
- Find out the data types of the columns to ensure that data is stored in the preferred format and the value of each property is as expected.
- Check the statistical summary of the dataset to get an overview of the numerical columns of the data.

**Observation 1:** The dataset has 495 rows and 19 columns.

**Observation 2:** All the columns in the data are non-null, except for 'entrepreneurs' and 'website'.

**Observation 3:** It's very crucial to identify the missing values, and treat them before moving ahead with the data analysis, and as checked there are missing values in the data.

**Observation 4:** There are 495 unique entrepreneur data. There are no duplicate values in the data. Deal column has two attributes 'True – secured a deal' and 'False – not secured a deal'.

**Observation 5:** Statistical summary is nothing but the technique to identify variables such as min, max, mean, standard deviation, count, 25th, 50th, & 75th percentile of the data.

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| episode | 495.0 | 1.212525e+01 | 7.918133e+00 | 1.0 | 5.0 | 11.0 | 18.0 | 29.0 |
| askedFor | 495.0 | 2.584909e+05 | 4.615999e+05 | 10000.0 | 75000.0 | 150000.0 | 250000.0 | 5000000.0 |
| exchangeForStake | 495.0 | 1.753939e+01 | 1.006105e+01 | 3.0 | 10.0 | 15.0 | 20.0 | 100.0 |
| valuation | 495.0 | 2.165615e+06 | 3.761971e+06 | 40000.0 | 440000.0 | 1000000.0 | 2000000.0 | 30000000.0 |
| season | 495.0 | 4.048485e+00 | 1.658518e+00 | 1.0 | 3.0 | 4.0 | 5.0 | 6.0 |

## Topic 1: Data Preparation and Exploratory Data Analysis:

**Question 1: Pick out the Deal (Dependent Variable) and Description columns into a separate data frame:**

| | deal | description |
|---|---|---|
| 0 | False | Bluetooth device implant for your ear. |
| 1 | True | Retail and wholesale pie factory with two reta... |
| 2 | True | Ava the Elephant is a godsend for frazzled par... |
| 3 | False | Organizing, packing, and moving services deliv... |
| 4 | False | Interactive media centers for healthcare waiti... |

**Question 2: Create two corpora (data frames which have description and deal columns) one with those who secured a Deal, the other with those who did not secure a deal:**

Corpora 1: deal secured

| | deal | description |
|---|---|---|
| 1 | True | Retail and wholesale pie factory with two reta... |
| 2 | True | Ava the Elephant is a godsend for frazzled par... |
| 5 | True | One of the first entrepreneurs to pitch on Sha... |
| 9 | True | An educational record label and publishing hou... |
| 10 | True | A battery-operated cooking device that siphons... |

Corpora 2: deal not secured

| | deal | description |
|---|---|---|
| 0 | False | Bluetooth device implant for your ear. |
| 3 | False | Organizing, packing, and moving services deliv... |
| 4 | False | Interactive media centers for healthcare waiti... |
| 6 | False | A mixed martial arts clothing line looking to ... |
| 7 | False | Attach Noted is a detachable "arm" that holds ... |

**Question 3: Find the number of characters in each row of description column for the corpus which secured the deal:**

```
1        73
2       244
5       365
9       122
10      117
```

**<u>Topic 2: Text Pre-processing:</u>** We'll be doing text preprocessing on the corpus for those who secured the deal.

**Step 1: Removal of http links:**

| | deal | description | char_count |
|---|---|---|---|
| 1 | True | Retail and wholesale pie factory with two reta... | 73 |
| 2 | True | Ava the Elephant is a godsend for frazzled par... | 244 |
| 5 | True | One of the first entrepreneurs to pitch on Sha... | 365 |
| 9 | True | An educational record label and publishing hou... | 122 |
| 10 | True | A battery-operated cooking device that siphons... | 117 |

a. **Pre-processing: De-contraction of words:**

| | deal | description | char_count |
|---|---|---|---|
| 1 | True | Retail and wholesale pie factory with two reta... | 73 |
| 2 | True | Ava the Elephant is a godsend for frazzled par... | 244 |
| 5 | True | One of the first entrepreneurs to pitch on Sha... | 365 |
| 9 | True | An educational record label and publishing hou... | 122 |
| 10 | True | A battery-operated cooking device that siphons... | 117 |

b. **Pre Processing : Removal of number:**

| | deal | description | char_count |
|---|---|---|---|
| 1 | True | Retail and wholesale pie factory with two reta... | 73 |
| 2 | True | Ava the Elephant is a godsend for frazzled par... | 244 |
| 5 | True | One of the first entrepreneurs to pitch on Sha... | 365 |
| 9 | True | An educational record label and publishing hou... | 122 |
| 10 | True | A battery-operated cooking device that siphons... | 117 |

**Step 2: Tokenization:** Tokenization is the process of breaking down a text into smaller units, typically words or tokens. These tokens are the building blocks of natural language processing (NLP) tasks such as text analysis, sentiment analysis, and machine translation.

| | deal | description | char_count |
|---|---|---|---|
| **1** | True | [Retail, and, wholesale, pie, factory, with, t... | 73 |
| **2** | True | [Ava, the, Elephant, is, a, godsend, for, fraz... | 244 |
| **5** | True | [One, of, the, first, entrepreneurs, to, pitch... | 365 |
| **9** | True | [An, educational, record, label, and, publishi... | 122 |
| **10** | True | [A, battery-operated, cooking, device, that, s... | 117 |

**Step 3: Lowercase:** Lowercasing ALL your text data, although commonly overlooked, is one of the simplest and most effective forms of text preprocessing. It is applicable to most text mining and NLP problems and can help in cases where your dataset is not very large and significantly helps with consistency of expected output.

```
1     [retail, and, wholesale, pie, factory, with, t...
2     [ava, the, elephant, is, a, godsend, for, fraz...
5     [one, of, the, first, entrepreneurs, to, pitch...
9     [an, educational, record, label, and, publishi...
10    [a, battery-operated, cooking, device, that, s...
```

**Step 4: Removal of punctuation:**

```
1     [retail, and, wholesale, pie, factory, with, t...
2     [ava, the, elephant, is, a, godsend, for, fraz...
5     [one, of, the, first, entrepreneurs, to, pitch...
9     [an, educational, record, label, and, publishi...
10    [a, batteryoperated, cooking, device, that, si...
```

**Step 5: Removal of stopwords:** Stop words are a set of commonly used words in a language. Examples of stop words in English are “a”, “the”, “is”, “are” and etc. The intuition behind using stop words is that, by removing low information words from text, we can focus on the important words instead.

```
1     [retail, wholesale, pie, factory, two, retail,...
2     [ava, elephant, godsend, frazzled, parents, yo...
5     [one, first, entrepreneurs, pitch, shark, tank...
9     [educational, record, label, publishing, house...
10    [batteryoperated, cooking, device, siphons, ju...
```

**Step 6: Lemmatization:** Lemmatization is particularly useful in natural language processing (NLP) tasks where it's important to accurately identify the meaning of words. By reducing words to their base forms, lemmatization helps improve the accuracy of tasks such as text classification, sentiment analysis, and information retrieval.

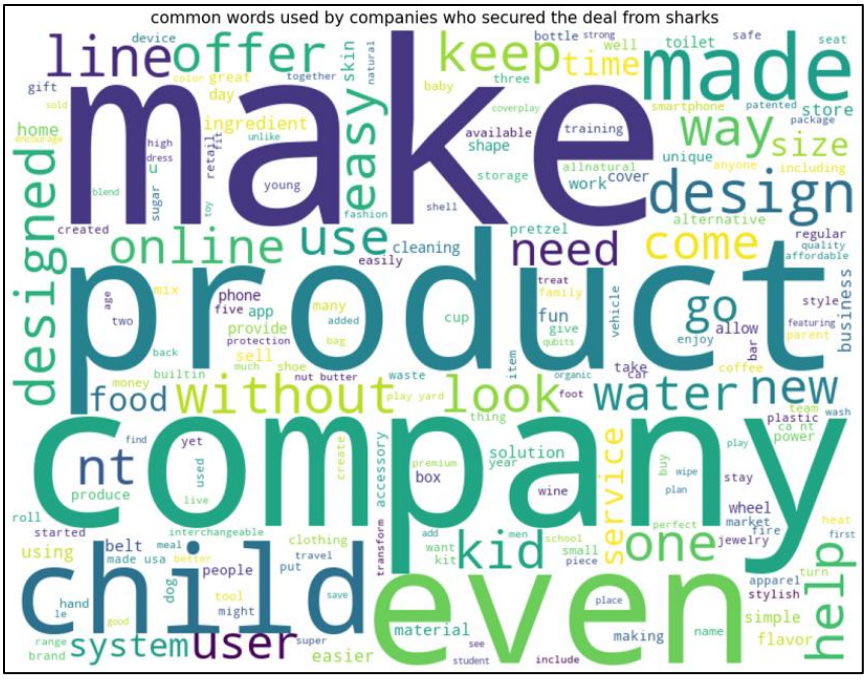
```
1     [retail, wholesale, pie, factory, two, retail,...
2     [ava, elephant, godsend, frazzled, parent, you...
5     [one, first, entrepreneur, pitch, shark, tank,...
9     [educational, record, label, publishing, house...
10    [batteryoperated, cooking, device, siphon, jui...
```

**Step 7: Text preprocessing-Normalization (aggregating pre-processing function into one):**

```
1     retail wholesale pie factory two retail locati...
2     ava elephant godsend frazzled parent young chi...
5     one first entrepreneur pitch shark tank susan ...
9     educational record label publishing house desi...
10    batteryoperated cooking device siphon juice si...
```

## Topic 3: Word Clouds:

- **Deal Secured(deal_yes):**


common words used by companies who secured the deal from sharks
make
product
company
child
even
line
offer
keep
made
way
size
design
easy
designed
online
use
need
come
go
without
look
water
new
food
nt
kid
one
help
system
user
service
solution
box
wine
material
easier
making
simple
flavor
apparel
stylish
wheel
market
jewelry
stay
plan
buy
wipe
clothing
travel
people
belt
using
started
roll
interchangeable
meal
made usa
dog
tool
might
hand
place
name
include
student
super
range
brand
see
transform
accessory
create
premium
add
want
kit
men
school
small
piece
perfect
fire
first
heat
turn
plastic
power
play
yard
foot
thing
wash
team
ca nt
business
style
featuring
parent
bar
coffee
take
car
item
vehicle
enjoy
allow
give
qubits
cup
fun
tree
family
phone
five
app
provide
protection
sell
much
shoe
builtin
nut butter
waste
organic
many
added
bag
two
age
mix
yet
find
used
live
produce
toilet
safe
seat
bottle
strong
well
time
three
together
great
day
gift
sold
home
high
dress
retail
fit
unlike
blend
sugar
toy
u
young
fashion
shell
created
easily
cleaning
pretzel
alternative
regular
quality
affordable
work
cover
allnatural
storage
shape
available
training
coverplay
smartphone
patented
store
package
including
anyone
unique
baby
skin
natural
ingredient
gift
color
day
together
great
three
coverplay
smartphone
patented
store
package
including
anyone
unique
size
way
design
alternative
regular
quality
affordable
business
style
featuring
parent
bar
coffee
take
car
item
vehicle
enjoy
allow
give
qubits
cup
fun
tree
family
phone
five
app
provide
protection
sell
much
shoe
builtin
nut butter
waste
organic
many
added
bag
two
age
mix
yet
find
used
live
produce
toilet
safe
seat
bottle
strong
well
time
three
together
great
day
gift
sold
home
high
dress
retail
fit
unlike
blend
sugar
toy
u
young
fashion
shell
created
easily
cleaning
pretzel
alternative

**Actionable Insights:**

- The **'Secured a deal'** word cloud contains words such as 'one', 'design', 'free', 'child', 'offer', 'easy' ,'online', 'use'.
- These indicate that Deals aimed towards catering to the children, who provided offers or a free sample/product, was easy to use, had a good design and was unique in its creativity are more likely to secure a deal.
- We could see that deals which are focused on children's use seem to secure a deal in most of the cases.
- The **'Did not secure a deal'** word cloud contains words such as 'one', 'designed' ,'help', 'device' ,'bottle', 'premium' ,'use', 'service'.
- These indicate that Deals with a mediocre design, less suited to solve/help a problem, products involving water bottles, having a higher and premium price tag and less usability are less likely to secure a deal.
- We could see that the service needs to be better in of the products which made the users not to secure a deal.
- It is also observed that words such as 'one', 'designed', 'system' and 'use' have a higher weight in both these word clouds.
- This indicates that either these were not the defining factors to whether a deal is made or not or might have been used in a different context in the description in each scenario.